

# Furniture.com – Data Scientist Take-home Case Study Guidelines

## Summary

Company X is a Real Estate aggregator company that lists properties across the country on their centralized platform. Property owners can list their properties on the platform, and interested parties can directly contact the owner if they like to enquire about the property.

## Problem

Users have noticed similar properties in similar areas can be significantly different in price at times. They have contacted the support team and raised concerns, and Company X is now experiencing a lack of trust in their platform.

## Purpose

Company X has now enlisted you to build a price discovery and regulation model that estimates the price range of a property given attributes like area, apartment type, amenities, etc.

## Project Specifications

- Create a price prediction model.
- Consider experimental design choices for the results to be valid in production.
- Provide clear visualizations to explain your results.
- Describe how you would use these results to recommend an action to the business.
- Please only spend 3-5 hours on this case study.

## Files to Submit

- 1 Jupyter Notebook
- Requirements.txt: we will use 'pip install -r requirements.txt' in a virtualenv
- 1 PowerPoint presentation

**Our team will review your case study and PowerPoint following submissions. If chosen for a final round, you will be asked to present your PowerPoint on your case study, and we will reference both your Jupyter Notebook and your presentation in the final rounds.**

## Data Description

The dataset contains information about 200 properties in Pune, Maharashtra, India with various attributes such as area, amenities, description, apartment type, etc.

See below for column information and details where needed:

- Sr. No: Identification Number
- Location
- Sub-Area: regions within the location
- Property Type: bhk refers to bedroom hall kitchen
- Property Area in Sq. Ft
- Price in Lakhs: lakhs refers to a unit in the India numbering system
- Price in Millions
- Company Name
- Township name/society name
- Total Township area in acres
- Club house
- School/university in township
- Hospital in township
- Mall in township
- Park / jogging track
- Swimming pool
- Gym
- Description: description of property